

Stochastic Integration: Intuition, Construction, and Applications

Lecture 3 Notes — Class Support and Student Summary

Goals and logical structure

The stochastic integral is built from left-point trading gains. The construction has three layers:

1. define the integral for simple predictable processes;
2. prove the Itô isometry;
3. extend by completion in $L^2(\Omega \times [0, T])$.

Under the expected finite-energy condition, the resulting gain process is a square-integrable martingale. Deterministic integrands then give explicit Gaussian applications, while the example $\int W \, dW$ reveals the role of Brownian quadratic variation. The final section translates the construction into reliable simulation rules.

1 Why Brownian Motion Needs a New Integral

1.1 From discrete trading gains to a continuous-time target

Fix a grid

$$0 = t_0 < t_1 < \cdots < t_n = T.$$

If ϕ_{t_i} is the number of shares selected at time t_i and held over $(t_i, t_{i+1}]$, the self-financing gain is

$$G_T^\pi = \sum_{i=0}^{n-1} \phi_{t_i} (S_{t_{i+1}} - S_{t_i}).$$

The continuous-time notation

$$\int_0^T \phi_t \, dS_t$$

should preserve this economic meaning: position chosen before a price move, multiplied by that fresh move, and then summed through time.

For a differentiable path S , ordinary calculus would give

$$\int_0^T \phi_t \, dS_t = \int_0^T \phi_t S'_t \, dt.$$

More generally, Riemann–Stieltjes integration works well when the integrator has finite variation. Brownian paths have neither property: almost surely they are nowhere differentiable and have infinite total variation on every nontrivial interval.

The scaling already explains the difficulty. Over an interval of length Δt ,

$$\Delta W \sim \mathcal{N}(0, \Delta t), \quad |\Delta W| \text{ is typically of order } \sqrt{\Delta t}.$$

On a uniform grid with n intervals, the sum of absolute increments is therefore heuristically of order

$$n\sqrt{T/n} = \sqrt{nT} \rightarrow \infty.$$

1.2 Quadratic variation is the surviving scale

For a partition π , define

$$Q_T^\pi(W) = \sum_i (W_{t_{i+1}} - W_{t_i})^2.$$

Theorem 1.1 (Quadratic variation on a uniform grid). If $t_i = iT/n$, then

$$Q_T^{(n)}(W) \xrightarrow[n \rightarrow \infty]{L^2} T.$$

In particular, the convergence holds in probability.

Proof. Write $\Delta W_i = W_{t_{i+1}} - W_{t_i}$ and $\Delta t = T/n$. The increments are independent and $\Delta W_i \sim \mathcal{N}(0, \Delta t)$. Thus

$$\mathbb{E}[Q_T^{(n)}] = \sum_i \mathbb{E}[(\Delta W_i)^2] = n\Delta t = T.$$

For a centered Gaussian variable with variance σ^2 , $\text{Var}(Z^2) = 2\sigma^4$. Hence

$$\text{Var}(Q_T^{(n)}) = \sum_i \text{Var}((\Delta W_i)^2) = 2n(\Delta t)^2 = \frac{2T^2}{n} \rightarrow 0.$$

Therefore $\mathbb{E}[(Q_T^{(n)} - T)^2] \rightarrow 0$. □

The key point is the change of scale:

$$\sum_i |\Delta W_i| \rightarrow \infty, \quad \sum_i (\Delta W_i)^2 \rightarrow T.$$

The second-order term does not disappear under refinement. It will create the correction terms characteristic of stochastic calculus.

1.3 The sampling convention matters

Take the random integrand $\phi_t = W_t$ and compare the left- and right-point sums

$$L^\pi = \sum_i W_{t_i} \Delta W_i, \quad R^\pi = \sum_i W_{t_{i+1}} \Delta W_i.$$

Since $W_{t_{i+1}} = W_{t_i} + \Delta W_i$,

$$R^\pi - L^\pi = \sum_i (\Delta W_i)^2 \rightarrow T.$$

Thus the two natural-looking conventions differ by an order-one quantity. There is no convention-free object called $\int W dW$. The Itô integral chooses left endpoints.

1.4 Filtration and predictability

Work on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$ supporting Brownian motion W . At time t_i , the trader knows $\mathcal{F}_{t_i}$. A legal position for the next interval must be $\mathcal{F}_{t_i}$ -measurable.

Definition 1.2 (Discrete predictability). The coefficient used on $(t_i, t_{i+1}]$ is predictable when it is determined by the information available at t_i . Equivalently, it may depend on the past and present, but not on $\Delta W_i = W_{t_{i+1}} - W_{t_i}$.

This is why left endpoints are not a cosmetic convention. They encode the information constraint of non-anticipative trading.

Exercise — which rules are admissible?

On the grid 0, 1, 2, decide whether each rule is predictable:

1. $\phi_0 = 1, \phi_1 = 1 + W_1^2$;
2. $\phi_0 = \text{sign}(W_1), \phi_1 = 0$;
3. $\phi_0 = 0, \phi_1 = \mathbf{1}_{\{W_1 > 0\}}$;
4. $\phi_i = \text{sign}(\Delta W_i)$.

Short solution

Rules 1 and 3 are admissible. In rule 1, $1 + W_1^2$ is known at time 1; in rule 3, $\{W_1 > 0\} \in \mathcal{F}_1$. Rule 2 uses W_1 at time 0, and rule 4 uses the future increment itself, so both anticipate.

If $\phi_{t_i} \in \mathcal{F}_{t_i}$ and the products are integrable, then

$$\mathbb{E}[\phi_{t_i} \Delta W_i \mid \mathcal{F}_{t_i}] = \phi_{t_i} \mathbb{E}[\Delta W_i \mid \mathcal{F}_{t_i}] = 0.$$

Consequently, $\mathbb{E}[G_T^\pi] = 0$. A predictable strategy can reshape risk, but cannot manufacture an average drift from Brownian increments.

2 Construction from Simple Predictable Processes

2.1 The integrability class

The intended integrands satisfy two conditions:

$$\phi \text{ is predictable,} \quad \mathbb{E} \int_0^T \phi_t^2 dt < \infty.$$

The second assumption is called the *expected finite-energy condition*. It means precisely that

$$\phi \in L^2(\Omega \times [0, T], \mathbb{P} \otimes dt).$$

The square is natural because a Brownian increment over length Δt has variance Δt : the contribution of a position ϕ to gain variance is therefore proportional to $\phi^2 \Delta t$.

Integrability is part of the definition, not an afterthought

Predictability controls information; square integrability controls size. Both are needed for the L^2 construction. Later, the same assumption will imply that the gain at every time belongs to $L^2 \subset L^1$, which is essential before one may call the gain process a martingale.

2.2 Definition for simple processes

Definition 2.1 (Simple predictable process). Fix $0 = t_0 < t_1 < \dots < t_n = T$. A simple predictable process has the form

$$\phi_t = \sum_{i=0}^{n-1} \xi_i \mathbf{1}_{(t_i, t_{i+1}]}(t), \quad \xi_i \in \mathcal{F}_{t_i},$$

with $\mathbb{E}[\xi_i^2] < \infty$. The coefficient ξ_i is selected at time t_i and held over the next interval.

Definition 2.2 (Itô integral of a simple process). For such a process, define

$$\int_0^T \phi_t dW_t := \sum_{i=0}^{n-1} \xi_i (W_{t_{i+1}} - W_{t_i}).$$

The definition is literally the discrete gain formula. It immediately gives the sanity checks

$$\int_0^T c dW_t = cW_T, \quad \int_0^T \mathbf{1}_{(a,b]}(t) dW_t = W_b - W_a,$$

and time additivity

$$\int_s^t \phi_r dW_r = \int_s^u \phi_r dW_r + \int_u^t \phi_r dW_r, \quad s \leq u \leq t.$$

The map is linear in the integrand. Processes that agree $(\mathbb{P} \otimes dt)$ -almost everywhere have the same integral almost surely.

Example — a deterministic step strategy

Let $T = 3$ and

$$\phi_t = 2\mathbf{1}_{(0,1]}(t) - \mathbf{1}_{(1,3]}(t).$$

Then

$$I = \int_0^3 \phi_t dW_t = 2W_1 - (W_3 - W_1).$$

The two increments are independent, so

$$I \sim \mathcal{N}(0, 4 \cdot 1 + 1 \cdot 2) = \mathcal{N}(0, 6).$$

Each interval contributes $(\text{position})^2 \times (\text{interval length})$ to the variance.

Exercise — switch after observing W_1

Let $A = \{W_1 > 0\}$ and

$$\phi_t = \mathbf{1}_A \mathbf{1}_{(1,2]}(t).$$

Explain why ϕ is predictable, write its integral without integral notation, compute its first two moments, and determine whether its law is Gaussian.

Short solution

Since $A \in \mathcal{F}_1$, the position is known before $(1, 2]$, and

$$I = \int_0^2 \phi_t dW_t = \mathbf{1}_A (W_2 - W_1).$$

The event A is independent of the fresh increment $W_2 - W_1 \sim \mathcal{N}(0, 1)$. Hence

$$\mathbb{E}[I] = 0, \quad \mathbb{E}[I^2] = \mathbb{P}(A) \mathbb{E}[(W_2 - W_1)^2] = \frac{1}{2}.$$

However, $\mathbb{P}(I = 0) = \mathbb{P}(A^c) = 1/2$. Thus

$$\mathcal{L}(I) = \frac{1}{2} \delta_0 + \frac{1}{2} \mathcal{N}(0, 1),$$

which is not Gaussian. Random predictable integrands do not generally produce Gaussian stochastic integrals.

2.3 The accumulated gain process

For a simple process, define

$$M_t = \int_0^t \phi_s dW_s.$$

On $(t_i, t_{i+1}]$,

$$M_t = M_{t_i} + \xi_i (W_t - W_{t_i}).$$

Thus the gain already accumulated is carried forward, while the current position multiplies the fresh Brownian move. This interpretation survives the L^2 extension below.

3 The Itô Isometry and the L^2 Extension

3.1 Orthogonality of fresh increments

Let $\Delta W_i = W_{t_{i+1}} - W_{t_i}$. Brownian increments on disjoint intervals are independent, so

$$\mathbb{E}[\Delta W_i \Delta W_j] = 0 \quad (i \neq j), \quad \mathbb{E}[(\Delta W_i)^2] = t_{i+1} - t_i.$$

For random predictable coefficients, the operational identity is

$$\mathbb{E}[\xi_i \Delta W_i \mid \mathcal{F}_{t_i}] = \xi_i \mathbb{E}[\Delta W_i \mid \mathcal{F}_{t_i}] = 0.$$

The later Brownian increment is fresh after conditioning on everything known before it. This conditional centering is what removes every off-diagonal term in the second-moment calculation.

Theorem 3.1 (Itô isometry for simple predictable processes). If ϕ is simple, predictable, and square-integrable, then

$$\mathbb{E} \left[\left(\int_0^T \phi_t dW_t \right)^2 \right] = \mathbb{E} \int_0^T \phi_t^2 dt.$$

Equivalently,

$$\left\| \int_0^T \phi_t dW_t \right\|_{L^2(\Omega)} = \|\phi\|_{L^2(\Omega \times [0, T])}.$$

Proof. Write $I = \sum_i \xi_i \Delta W_i$. Expanding the square gives

$$\mathbb{E}[I^2] = \sum_i \mathbb{E}[\xi_i^2 (\Delta W_i)^2] + 2 \sum_{i < j} \mathbb{E}[\xi_i \Delta W_i \xi_j \Delta W_j].$$

If $i < j$, then $\xi_i \Delta W_i \xi_j$ is $\mathcal{F}_{t_j}$ -measurable. Hence

$$\begin{aligned} \mathbb{E}[\xi_i \Delta W_i \xi_j \Delta W_j] &= \mathbb{E}[\xi_i \Delta W_i \xi_j \mathbb{E}[\Delta W_j \mid \mathcal{F}_{t_j}]] \\ &= 0. \end{aligned}$$

On the diagonal, ΔW_i is independent of $\mathcal{F}_{t_i}$, so

$$\mathbb{E}[\xi_i^2 (\Delta W_i)^2] = \mathbb{E}[\xi_i^2] (t_{i+1} - t_i).$$

Therefore

$$\mathbb{E}[I^2] = \sum_i \mathbb{E}[\xi_i^2] (t_{i+1} - t_i) = \mathbb{E} \int_0^T \phi_t^2 dt.$$

□

Geometric interpretation

The stochastic integral is an isometry: it transforms an exposure profile in $L^2(\Omega \times [0, T])$ into a terminal random gain in $L^2(\Omega)$ without changing its squared norm. Integrand energy becomes gain variance.

3.2 Mean, covariance, and risk bounds

The same conditioning argument gives

$$\mathbb{E} \left[\int_0^T \phi_t dW_t \right] = 0.$$

Applying the isometry to $\phi + \psi$ and $\phi - \psi$, or expanding directly, gives the polarization identity.

Proposition 3.2 (Covariance identity). For square-integrable predictable ϕ and ψ ,

$$\mathbb{E} \left[\left(\int_0^T \phi_t dW_t \right) \left(\int_0^T \psi_t dW_t \right) \right] = \mathbb{E} \int_0^T \phi_t \psi_t dt.$$

Define the pathwise exposure and its expected energy by

$$\mathcal{E}_T(\phi) = \int_0^T \phi_t^2 dt, \quad \mathbb{E}[\mathcal{E}_T(\phi)] = \mathbb{E} \int_0^T \phi_t^2 dt.$$

The isometry implies:

- multiplying the position by c multiplies gain variance by c^2 ;
- holding a fixed position twice as long doubles its variance contribution;
- moving exposure across time may preserve terminal variance while changing the whole gain path.

If $|\phi_t| \leq L$, then

$$\text{Var} \left(\int_0^T \phi_t dW_t \right) = \mathbb{E} \int_0^T \phi_t^2 dt \leq L^2 T.$$

Exercise — same terminal risk, different timing

On $[0, 1]$, compare

$$\phi_t = 1, \quad \psi_t = \sqrt{2} \mathbf{1}_{(0, 1/2]}(t).$$

Compute their energies and stochastic integrals. Compare their terminal laws and their gain processes before time 1.

Short solution

Both strategies have unit energy:

$$\int_0^1 \phi_t^2 dt = 1, \quad \int_0^1 \psi_t^2 dt = 2 \cdot \frac{1}{2} = 1.$$

Moreover,

$$\int_0^1 \phi_t dW_t = W_1, \quad \int_0^1 \psi_t dW_t = \sqrt{2} W_{1/2}.$$

Both terminal gains are $\mathcal{N}(0, 1)$. Their paths are not identical: the second gain stops changing after time 1/2, while the first remains exposed until time 1.

3.3 Completion in L^2

Let ϕ be predictable with

$$\mathbb{E} \int_0^T \phi_t^2 dt < \infty.$$

Choose simple predictable processes ϕ^n such that

$$\mathbb{E} \int_0^T |\phi_t^n - \phi_t|^2 dt \longrightarrow 0.$$

The isometry gives

$$\mathbb{E} \left[\left| \int_0^T \phi_t^n dW_t - \int_0^T \phi_t^m dW_t \right|^2 \right] = \mathbb{E} \int_0^T |\phi_t^n - \phi_t^m|^2 dt.$$

The right-hand side tends to zero, so the sequence of integrals is Cauchy in the complete space $L^2(\Omega)$.

Definition 3.3 (General Itô integral). For a square-integrable predictable process ϕ , define

$$\int_0^T \phi_t dW_t := L^2\text{-}\lim_{n \rightarrow \infty} \int_0^T \phi_t^n dW_t.$$

The definition does not depend on the approximating sequence. Indeed, if $\psi^n \rightarrow \phi$ as well, the isometry yields

$$\mathbb{E} \left[\left| \int_0^T \phi_t^n dW_t - \int_0^T \psi_t^n dW_t \right|^2 \right] = \mathbb{E} \int_0^T |\phi_t^n - \psi_t^n|^2 dt \rightarrow 0.$$

Linearity, zero mean, the isometry, and the covariance identity all pass to the limit.

The construction in one sentence

Approximate the predictable exposure, integrate the simple approximations, and take the L^2 limit. The isometry simultaneously proves convergence, uniqueness, and stability.

4 The Integral as a Martingale Gain Process

For a predictable process ϕ satisfying

$$\mathbb{E} \int_0^T \phi_s^2 ds < \infty,$$

define the accumulated gain

$$M_t = \int_0^t \phi_s dW_s, \quad 0 \leq t \leq T.$$

4.1 The integrability audit

Before calling M a martingale, three points must be checked:

1. M_t is $\mathcal{F}_t$ -measurable;
2. $M_t \in L^1$ for every t ;
3. $\mathbb{E}[M_t | \mathcal{F}_s] = M_s$ for $s \leq t$.

Adaptedness follows first for simple integrands and then by approximation. The second point is exactly where the expected finite-energy condition matters. The isometry and Cauchy–Schwarz give

$$\begin{aligned} \mathbb{E}|M_t| &\leq (\mathbb{E}[M_t^2])^{1/2} \\ &= \left(\mathbb{E} \int_0^t \phi_s^2 ds \right)^{1/2} < \infty. \end{aligned}$$

Thus $M_t \in L^2 \subset L^1$. In particular, the conditional expectation $\mathbb{E}[M_t | \mathcal{F}_s]$ is well defined.

True martingale versus local martingale

There are two distinct assumptions:

$$\mathbb{E} \int_0^T \phi_s^2 ds < \infty \quad \text{and} \quad \int_0^T \phi_s^2 ds < \infty \quad \text{a.s.}$$

The first gives the square-integrable, hence true, martingale used throughout this lecture. If only the second holds locally on every finite horizon, the stochastic integral is in general only a continuous local martingale. A local martingale need not be integrable at every time and need not preserve its expectation. One must never silently replace the expected L^2 condition by pathwise finiteness.

4.2 Martingale property and proof

Theorem 4.1 (Square-integrable martingale property). If ϕ is predictable and $\mathbb{E} \int_0^T \phi_s^2 ds < \infty$, then

$$M_t = \int_0^t \phi_s dW_s$$

has a continuous adapted version, $M_t \in L^2$, and

$$\boxed{\mathbb{E}[M_t | \mathcal{F}_s] = M_s, \quad 0 \leq s \leq t \leq T.}$$

Equivalently,

$$\mathbb{E} \left[\int_s^t \phi_u dW_u \middle| \mathcal{F}_s \right] = 0.$$

Proof. First suppose that ϕ is simple and use a grid containing s and t . Splitting past and future gains gives

$$M_t = M_s + \sum_{t_i \geq s} \xi_i \Delta W_i.$$

For each future term,

$$\mathbb{E}[\xi_i \Delta W_i | \mathcal{F}_{t_i}] = \xi_i \mathbb{E}[\Delta W_i | \mathcal{F}_{t_i}] = 0.$$

Successive applications of the tower property yield

$$\mathbb{E}[M_t - M_s | \mathcal{F}_s] = 0.$$

For a general integrand, choose simple $\phi^n \rightarrow \phi$ in $L^2(\Omega \times [0, T])$, and let $M_t^n = \int_0^t \phi_u^n dW_u$. The isometry gives $M_t^n \rightarrow M_t$ in L^2 , hence in L^1 . Conditional expectation is an L^1 -contraction:

$$\mathbb{E} |\mathbb{E}[M_t^n - M_t | \mathcal{F}_s]| \leq \mathbb{E} |M_t^n - M_t| \rightarrow 0.$$

Therefore one may pass to the limit in $\mathbb{E}[M_t^n | \mathcal{F}_s] = M_s^n$, proving the claim. The continuous version follows from the standard construction of the Brownian stochastic integral. $\square$

The proof contains the reusable mechanism of stochastic integration:

$$\text{predictable coefficient} \times \text{fresh centered increment} \implies \text{zero conditional mean.}$$

4.3 Conditional variance and state-dependent risk

The conditional version of the isometry is

$$\boxed{\mathbb{E} \left[\left(\int_s^t \phi_u dW_u \right)^2 \middle| \mathcal{F}_s \right] = \mathbb{E} \left[\int_s^t \phi_u^2 du \middle| \mathcal{F}_s \right].}$$

Because the future integral has conditional mean zero, this identity is also its conditional variance. The mean forecast is unchanged, while the amount of risk still ahead may depend strongly on the current state.

Exercise — a state-dependent position

On $[0, 2]$, define

$$\phi_t = \mathbf{1}_{(0,1]}(t) + (1 + \mathbf{1}_{\{W_1 > 0\}}) \mathbf{1}_{(1,2]}(t), \quad M_t = \int_0^t \phi_s dW_s.$$

Compute M_1 , $\mathbb{E}[M_2 | \mathcal{F}_1]$, and $\text{Var}(M_2 - M_1 | \mathcal{F}_1)$.

Short solution

Up to time 1, the position is one, so $M_1 = W_1$. Since M is a martingale,

$$\mathbb{E}[M_2 \mid \mathcal{F}_1] = M_1 = W_1.$$

Over $(1, 2]$, the position is already known at time 1. Therefore

$$\begin{aligned} \text{Var}(M_2 - M_1 \mid \mathcal{F}_1) &= (1 + \mathbf{1}_{\{W_1 > 0\}})^2 \\ &= \begin{cases} 1, & W_1 \leq 0, \\ 4, & W_1 > 0. \end{cases} \end{aligned}$$

Adaptation changes conditional variance without creating conditional drift.

4.4 Quadratic variation as a random clock

Proposition 4.2 (Quadratic variation of a stochastic integral). For

$$M_t = \int_0^t \phi_s \, dW_s,$$

one has

$$[M]_t = \int_0^t \phi_s^2 \, ds.$$

For a simple strategy, the formula is already visible from

$$\sum_i (\Delta M_i)^2 = \sum_i \xi_i^2 (\Delta W_i)^2 \approx \sum_i \xi_i^2 \Delta t_i.$$

The process M therefore runs on the exposure clock generated by ϕ^2 . Calendar time advances uniformly, while the stochastic gain accumulates variation faster when the absolute position is larger.

4.5 Stopping an exposure

Let τ be a stopping time. The predictable strategy $\phi_s = \mathbf{1}_{\{s \leq \tau\}}$ holds one unit until τ and zero afterward. For every finite t ,

$$\int_0^t \mathbf{1}_{\{s \leq \tau\}} \, dW_s = W_{t \wedge \tau}.$$

Crucially, its integrability is automatic on a bounded horizon:

$$\mathbb{E} \int_0^t \mathbf{1}_{\{s \leq \tau\}}^2 \, ds = \mathbb{E}[t \wedge \tau] \leq t < \infty.$$

The martingale property and isometry may therefore be applied legitimately:

$$\mathbb{E}[W_{t \wedge \tau}] = 0, \quad \mathbb{E}[W_{t \wedge \tau}^2] = \mathbb{E}[t \wedge \tau].$$

Example — stopping at an upper barrier

Let

$$\tau_a = \inf\{t \geq 0 : W_t = a\}, \quad a > 0.$$

The stopped exposure is

$$M_t = \int_0^t \mathbf{1}_{\{s \leq \tau_a\}} \, dW_s = W_{t \wedge \tau_a}.$$

Without using the density of τ_a ,

$$\mathbb{E}[M_t] = 0, \quad \mathbb{E}[M_t^2] = \mathbb{E}[t \wedge \tau_a].$$

The first statement comes from the true-martingale property; the second comes from the isometry.

What predictability and integrability buy us

For every predictable ϕ satisfying $\mathbb{E} \int_0^T \phi_t^2 dt < \infty$, the gain process is a square-integrable martingale and

$$\mathbb{E} \left[\int_0^T \phi_t dW_t \right] = 0, \quad \mathbb{E} \left[\left(\int_0^T \phi_t dW_t \right)^2 \right] = \mathbb{E} \int_0^T \phi_t^2 dt.$$

The distribution may nevertheless be Gaussian, a mixture, stopped, skewed, or state-dependent. Zero conditional drift does not mean zero risk.

5 Deterministic Integrands and Gaussian Applications

5.1 Gaussianity and the L^2 inner product

When the integrand is deterministic, the stochastic integral has additional structure.

Theorem 5.1 (Distribution for a deterministic integrand). If $f \in L^2(0, T)$ is deterministic, then

$$I_f = \int_0^T f(t) dW_t \sim \mathcal{N} \left(0, \int_0^T f(t)^2 dt \right).$$

Proof. For a deterministic step function, the integral is a linear combination of independent Gaussian increments and is therefore Gaussian. Choose deterministic step functions $f_n \rightarrow f$ in $L^2(0, T)$. The isometry gives

$$\int_0^T f_n(t) dW_t \longrightarrow \int_0^T f(t) dW_t \quad \text{in } L^2(\Omega),$$

and the variances converge to $\int_0^T f(t)^2 dt$. The characteristic functions consequently converge to that of the stated centered Gaussian law. $\square$

The same reasoning for linear combinations gives joint Gaussianity.

Proposition 5.2 (Covariance and independence). For deterministic $f, g \in L^2(0, T)$,

$$\text{Cov} \left(\int_0^T f dW, \int_0^T g dW \right) = \int_0^T f(t)g(t) dt.$$

The pair is jointly Gaussian. Therefore

$$\int_0^T f(t)g(t) dt = 0 \implies \int_0^T f dW \text{ and } \int_0^T g dW \text{ are independent.}$$

The map

$$f \mapsto \int_0^T f dW$$

turns the deterministic $L^2(0, T)$ inner product into covariance. Ordinary orthogonality of exposure profiles becomes independence of Gaussian risks.

Example — a linearly increasing exposure

Take $f(t) = t$. Then

$$I_T = \int_0^T t \, dW_t$$

is centered Gaussian with

$$\text{Var}(I_T) = \int_0^T t^2 \, dt = \frac{T^3}{3}.$$

Thus

$$I_T \sim \mathcal{N}\left(0, \frac{T^3}{3}\right).$$

The standard deviation scales like $T^{3/2}$, which is consistent with a position of order T multiplied by Brownian noise of order $T^{1/2}$.

5.2 Conditioning as orthogonal projection

Let

$$I_T = \int_0^T t \, dW_t.$$

Since

$$W_T = \int_0^T 1 \, dW_t,$$

the covariance identity gives

$$\text{Cov}(I_T, W_T) = \int_0^T t \, dt = \frac{T^2}{2}.$$

For a centered jointly Gaussian pair (X, Y) ,

$$\mathbb{E}[X \mid Y] = \frac{\text{Cov}(X, Y)}{\text{Var}(Y)} Y, \quad \text{Var}(X \mid Y) = \text{Var}(X) - \frac{\text{Cov}(X, Y)^2}{\text{Var}(Y)}.$$

Using $\text{Var}(W_T) = T$, we obtain

$$\mathbb{E}[I_T \mid W_T] = \frac{T}{2} W_T$$

and

$$\text{Var}(I_T \mid W_T) = \frac{T^3}{3} - \frac{(T^2/2)^2}{T} = \frac{T^3}{12}.$$

Geometrically, conditioning on W_T reveals the component of $t \mapsto t$ parallel to the constant function 1. Indeed,

$$t = \frac{T}{2} + \left(t - \frac{T}{2}\right), \quad \int_0^T \left(t - \frac{T}{2}\right) \, dt = 0.$$

Exercise — construct a Gaussian risk independent of W_T

Find a nonzero affine function $g(t) = t - c$ such that

$$Y = \int_0^T g(t) \, dW_t$$

is independent of W_T . Compute $\text{Var}(Y)$.

Short solution

Because $W_T = \int_0^T 1 \, dW_t$, independence is equivalent to

$$0 = \text{Cov}(Y, W_T) = \int_0^T g(t) \, dt.$$

For $g(t) = t - c$, this gives $T^2/2 - cT = 0$, hence $c = T/2$. Therefore

$$g(t) = t - \frac{T}{2}, \quad \text{Var}(Y) = \int_0^T \left(t - \frac{T}{2}\right)^2 \, dt = \frac{T^3}{12}.$$

The pair is jointly Gaussian and uncorrelated, so it is independent.

5.3 Deterministic exposure as a time change

For deterministic f , define

$$M_t = \int_0^t f(s) \, dW_s, \quad q(t) = \int_0^t f(s)^2 \, ds.$$

Then M is centered Gaussian. For $s < t$,

$$M_t - M_s = \int_s^t f(u) \, dW_u$$

depends only on Brownian increments over $(s, t]$, so disjoint increments of M are independent. Moreover,

$$\text{Var}(M_t - M_s) = q(t) - q(s).$$

Consequently,

$$(M_t)_{t \geq 0} \stackrel{\text{law}}{=} (B_{q(t)})_{t \geq 0}$$

for a Brownian motion B . For $f(t) = t$, the stochastic clock is $q(t) = t^3/3$.

This time-change statement is special to deterministic integrands. For a random integrand, $\int_0^t \phi_s^2 \, ds$ is a random clock, but independent increments and Gaussianity generally disappear.

5.4 Integration by parts with a deterministic integrand

The useful integration-by-parts formula in this lecture does not require Itô's formula.

Proposition 5.3 (Deterministic integration by parts). Let $f : [0, T] \rightarrow \mathbb{R}$ be deterministic and continuously differentiable. More generally, assume that f is absolutely continuous with $f' \in L^2(0, T)$. Then

$$f(T)W_T - f(0)W_0 = \int_0^T f(t) \, dW_t + \int_0^T W_t f'(t) \, dt.$$

Since $W_0 = 0$,

$$\int_0^T f(t) \, dW_t = f(T)W_T - \int_0^T W_t f'(t) \, dt.$$

Proof. The assumption $f' \in L^2(0, T)$ implies $f' \in L^1(0, T)$ on the finite interval. Thus f has finite variation and belongs to $L^2(0, T)$. On a partition, discrete summation by parts gives

$$f(T)W_T - f(0)W_0 = \sum_i f(t_i) \Delta W_i + \sum_i W_{t_{i+1}} (f(t_{i+1}) - f(t_i)).$$

As the mesh tends to zero, the first sum converges in L^2 to $\int_0^T f(t) \, dW_t$. Since W is continuous and f has finite variation, the second converges pathwise to

$$\int_0^T W_t \, df(t) = \int_0^T W_t f'(t) \, dt.$$

This proves the formula directly from summation by parts. $\square$

Remark 5.4 (Why there is no correction term). The deterministic path f has finite variation, so its quadratic covariation with W is zero. No Itô formula is needed for this result.

Example — the Brownian area

Set

$$A_T = \int_0^T W_t \, dt.$$

Apply deterministic integration by parts with $f(t) = t$:

$$A_T = TW_T - \int_0^T t \, dW_t.$$

The right-hand side is a linear combination of jointly Gaussian stochastic integrals, so A_T is centered Gaussian. Using

$$\text{Var}(W_T) = T, \quad \text{Var}\left(\int_0^T t \, dW_t\right) = \frac{T^3}{3}, \quad \text{Cov}\left(W_T, \int_0^T t \, dW_t\right) = \frac{T^2}{2},$$

we find

$$\begin{aligned} \text{Var}(A_T) &= T^2 \text{Var}(W_T) + \text{Var}\left(\int_0^T t \, dW_t\right) - 2T \text{Cov}\left(W_T, \int_0^T t \, dW_t\right) \\ &= T^3 + \frac{T^3}{3} - T^3 = \frac{T^3}{3}. \end{aligned}$$

Therefore

$$A_T \sim \mathcal{N}\left(0, \frac{T^3}{3}\right).$$

6 Left, Right, and Midpoint Conventions

6.1 One integrand, three sums

For a partition π and the integrand $\phi_t = W_t$, define

$$\begin{aligned} L^\pi &= \sum_i W_{t_i} \Delta W_i, \\ M^\pi &= \sum_i \frac{W_{t_i} + W_{t_{i+1}}}{2} \Delta W_i, \\ R^\pi &= \sum_i W_{t_{i+1}} \Delta W_i. \end{aligned}$$

Their information content is different:

Sum	Evaluation rule	Interpretation
left	value known before the move	Itô
midpoint	uses both endpoints	Stratonovich
right	value known after the move	anticipative here

The exact identity

$$W_{t_{i+1}}^2 - W_{t_i}^2 = 2W_{t_i} \Delta W_i + (\Delta W_i)^2$$

telescopes to

$$W_T^2 = 2L^\pi + \sum_i (\Delta W_i)^2.$$

Since Brownian quadratic variation converges to T ,

$$\int_0^T W_t dW_t = \lim_{|\pi| \rightarrow 0} L^\pi = \frac{1}{2}(W_T^2 - T).$$

This is the first nontrivial stochastic integral. The term $-T/2$ is not obtained from the general Itô formula here; it follows directly from the telescoping identity and quadratic variation.

Furthermore,

$$R^\pi = L^\pi + \sum_i (\Delta W_i)^2, \quad M^\pi = \frac{1}{2}(L^\pi + R^\pi).$$

Therefore the three limits are

Convention	Limit
left / Itô	$\frac{1}{2}(W_T^2 - T)$
midpoint / Stratonovich	$\frac{1}{2}W_T^2$
right	$\frac{1}{2}(W_T^2 + T)$

Exercise — verify the isometry for $\int W dW$

Let

$$I_T = \int_0^T W_t dW_t = \frac{1}{2}(W_T^2 - T).$$

Compute $\mathbb{E}[I_T]$ and $\mathbb{E}[I_T^2]$, using $\mathbb{E}[W_T^4] = 3T^2$. Check the answer with the Itô isometry.

Short solution

Since $\mathbb{E}[W_T^2] = T$,

$$\mathbb{E}[I_T] = \frac{1}{2}(\mathbb{E}[W_T^2] - T) = 0.$$

Moreover,

$$\mathbb{E}[I_T^2] = \frac{1}{4}\mathbb{E}[(W_T^2 - T)^2] = \frac{1}{4}(3T^2 - 2T^2 + T^2) = \frac{T^2}{2}.$$

The isometry gives exactly the same result:

$$\mathbb{E}[I_T^2] = \mathbb{E} \int_0^T W_t^2 dt = \int_0^T \mathbb{E}[W_t^2] dt = \int_0^T t dt = \frac{T^2}{2}.$$

6.2 Why anticipation creates a fake arbitrage

Suppose illegally that one could choose

$$\phi_{t_i} = \text{sign}(\Delta W_i)$$

before reporting the gain. Then

$$\sum_i \phi_{t_i} \Delta W_i = \sum_i |\Delta W_i| \geq 0,$$

and the gain is strictly positive almost surely on every nontrivial grid. Under grid refinement, this sum even reflects the infinite total variation of Brownian paths.

The apparent opportunity comes entirely from hindsight: $\phi_{t_i} \notin \mathcal{F}_{t_i}$. Predictability is therefore both the mathematical boundary of the Itô integral and the economic boundary of a legal self-financing strategy.

7 Simulation of Stochastic Integrals

7.1 The left-point algorithm

Let $t_i = i\Delta t$, where $\Delta t = T/n$. Generate independent standard normal variables $Z_0, \dots, Z_{n-1}$ and set

$$\Delta W_i = \sqrt{\Delta t} Z_i.$$

Starting from $W_0 = 0$ and $I_0 = 0$, repeat for $i = 0, \dots, n-1$:

1. compute ϕ_i from information available at t_i ;
2. set $I_{i+1} = I_i + \phi_i \Delta W_i$;
3. set $W_{t_{i+1}} = W_{t_i} + \Delta W_i$.

Then

$$I_n = \sum_{i=0}^{n-1} \phi_i \Delta W_i$$

is the left-point approximation of $\int_0^T \phi_t dW_t$.

The update order encodes the filtration

The value ϕ_i must be fully determined before ΔW_i is used. If it is computed from $W_{t_{i+1}}$, or from the newly generated increment, the scheme is a right-point or otherwise anticipative sum rather than an Itô approximation.

For a target integrand ϕ , define the left-step approximation

$$\phi_t^{(n)} = \sum_{i=0}^{n-1} \phi_{t_i} \mathbf{1}_{(t_i, t_{i+1}]}(t).$$

Whenever $\phi^{(n)} \rightarrow \phi$ in $L^2(\Omega \times [0, T])$, the isometry gives the exact mean-square error identity

$$\mathbb{E} \left[\left| \int_0^T \phi_t dW_t - \sum_{i=0}^{n-1} \phi_{t_i} \Delta W_i \right|^2 \right] = \mathbb{E} \int_0^T |\phi_t - \phi_t^{(n)}|^2 dt.$$

Thus the numerical approximation is controlled by approximation of the integrand in the same L^2 norm used in the theoretical construction.

7.2 An exact benchmark

For

$$I = \int_0^T W_t dW_t, \quad I^{(n)} = \sum_{i=0}^{n-1} W_{t_i} \Delta W_i,$$

the identity $I = \frac{1}{2}(W_T^2 - T)$ gives

$$I - I^{(n)} = \frac{1}{2} \sum_{i=0}^{n-1} ((\Delta W_i)^2 - \Delta t).$$

For a uniform grid,

$$\text{Var}((\Delta W_i)^2) = 2(\Delta t)^2.$$

The centered summands are independent, so

$$\begin{aligned} \mathbb{E}[(I - I^{(n)})^2] &= \frac{1}{4} \sum_{i=0}^{n-1} 2(\Delta t)^2 \\ &= \frac{1}{2} n \left(\frac{T}{n} \right)^2 = \boxed{\frac{T^2}{2n}}. \end{aligned}$$

The root-mean-square error is therefore

$$\text{RMSE}(I^{(n)}) = \frac{T}{\sqrt{2n}}.$$

This supplies a clean convergence-rate test for an implementation.

7.3 Diagnostics and classic traps

Mistake	Consequence	Check or correction
Use $\Delta t Z_i$ for ΔW_i	Brownian variance is too small and has the wrong time scaling	Verify $\Delta W_i = \sqrt{\Delta t} Z_i$ and empirical variance Δt
Compute ϕ_i from $W_{t_{i+1}}$	The scheme anticipates and can introduce an order-one bias	Freeze ϕ_i from the state at t_i
Use fresh independent noise for a jointly driven integrand	The covariance structure and joint law are changed	Reuse the same Brownian increments whenever the model specifies the same driver
Replace ΔW_i by Δt	A stochastic integral is replaced by an ordinary time integral	Track dW and dt terms separately
Check only the empirical mean	Variance or dependence errors remain invisible	Test the isometry, covariance, and several grid sizes

A single plausible-looking path is not validation. Useful checks include:

- sample mean close to zero when the hypotheses imply a true martingale;
- sample second moment close to $\mathbb{E} \int_0^T \phi_t^2 dt$;
- covariance checks when several integrals use the same Brownian driver;
- error decay under successive grid refinements;
- comparison with an exact identity when one is available.

Final checklist

1. The Itô integral is the L^2 limit of left-point, non-anticipative gains.
2. Predictability controls information; expected finite energy controls integrability.
3. Under $\mathbb{E} \int_0^T \phi_t^2 dt < \infty$, the gain process is a square-integrable martingale with quadratic variation $\int_0^t \phi_s^2 ds$.
4. Deterministic integrands produce jointly Gaussian variables, and L^2 inner products become covariances.
5. Deterministic integration by parts follows from summation by parts and requires no Itô formula.
6. Brownian quadratic variation produces $\int_0^T W_t dW_t = \frac{1}{2}(W_T^2 - T)$.
7. A correct simulation respects the same filtration and L^2 structure as the theoretical integral.